

The Ultimate Representation of The Periodic Table

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What does the periodic table of elements actually tell us? NOTHING. At least, that's the conclusion reached by the authors of this paper. In its current form, Mendeleev's table is little more than a mosaic of colorful tiles. When confronted with it, part of the scientific community either improvises the physicochemical properties of an element based on its position among neighboring tiles (74% of respondents), while the rest rely on Tarot reading (40% of respondents [14% claim to use Tarot but are in fact just making things up, and 20% have only ever drawn The Fool card]). In this paper, we present a groundbreaking approach to elemental data representation: the traditional Mendeleev's table is dethroned and replaced by the Chernoff table. The key physicochemical parameters are encoded into the faces of elemental personas, allowing their properties to be grasped at a single glance.

Keywords: launcing; bouncing; Chernoff faces; periodic table; elements.

I. INTRODUCTION

In his 1973 paper [1], Herman Chernoff presented the world with what can only be described as a visualization Holy Grail: human faces. His idea was brilliantly simple and slightly mischievous. Each point in a k -dimensional space (originally for $k \leq 18$) is represented as a cartoon face. Individual facial features—such as the length of the nose, the curvature of the mouth, the size of the eyes, or the angle of the eyebrows—correspond directly to individual components of the data vector.

In other words, every multivariate observation becomes a computer-drawn face whose geometry encodes numerical values. Instead of staring at columns of numbers or sterile scatterplots, the observer is confronted with a gallery of faces. And because the human brain is exceptionally good at detecting subtle differences in facial features, patterns, clusters, regularities, and outliers in high-dimensional data suddenly become almost

intuitive. A slight smirk here, an unusually long nose there—and boom, your anomaly just revealed itself. Chernoff also briefly described alternative graphical representations, but let's be honest: once you realize you can turn data into faces, everything else feels like statistical small talk.

And since we're already on the subject of small talk—the periodic table of elements. It is the most popular way of presenting elemental data in the world. It was invented by Dmitri Mendeleev in 1869, while drinking vodka alone in some extremely cold place (the authors' speculation, strongly supported by life experience—because let's be honest, it takes a rather uneventful life to stare at a list of elements for hours, and a certain drunken imagination to arrange them into such zigzags).



FIG. 1: Left: Herman Chernoff, photograph from 2008. Right: one of the faces from Chernoff's original paper [1].

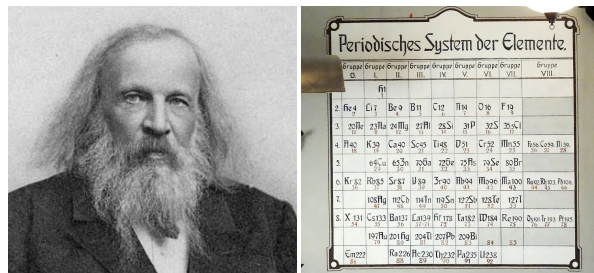


FIG. 2: Left: Dmitri Mendeleev, photograph from 1890. Right: Restored periodic table from Mendeleev's era on the wall of the Chemical Auditorium at Gdańsk University of Technology.

This table brought enormous benefits to humanity. Suddenly, understanding nature became just a little bit easier again. Over time, more and more rules emerged describing how the physicochemical properties of elements correlate with their position in the table. And so today, after completing a course in inorganic chemistry, every student and researcher is supposed to

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know (lmao) how the number of valence electrons, the number of shells, atomic radius, ionization energy, and so on and so forth... relate to the position of a given element in the table.

But the world is changing, and we are no longer capable of remembering everything. These days, out of sheer laziness, instead of opening Wikipedia to check the melting point of tritium, we type a prompt into an AI chat and then spend the rest of the day wondering whether it told the truth or completely screwed up the units (well, too late, the paper is already with the reviewer). Which is precisely why we need a new, condensed form of presenting elemental data.

The solution to this problem had been lying right in front of us since 1978, yet it took us nearly 50 years to actually see it.

Anime faces.

By exploiting Chernoff's idea and expanding it with the immaculately beautiful facial features of anime characters, we can finally upgrade Mendeleev's masterpiece and present The Ultimate Representation of the Periodic Table. The most important elemental properties (carefully selected to support our thesis, obviously) have been meticulously encoded as features of anime-style faces.

Now anyone can confidently print such a table to cover an entire office wall—and a single glance at Zinc-san is enough to estimate the approximate how easily it melts just from the intensity of the blush on his cheeks.

II. METHODS

The anime-stylized faces were drawn using Inkscape vector graphic suite, by one of the authors, based on their previous education in computer graphics and usage of online tutorials [2]. To allow for dynamic modification of the prepared faces, custom scripts were implemented using Python language with svgpathtools [3]. These scripts allow for:

- **Toggling** - changing visibility of selected paths in the drawing
- **Transforming** - scaling selected path with selected anchor point, as well as rotation around center of the path
- **Path modification** - changing selected line in a path to several connected b-splines that approximate given function

- **Coloring** - changing paths fill color, transparency or stroke properties

The function used for hairline calculation [4], which we decided to call Norwood function, is defined as:

$$N(x) = -p(x/e)^4 + -p(x/e)^2 - f \quad (1)$$

where p is the prominence of receding hairline, e and f are coefficients allowing for good placement on the character head. After experimentation with parameters $e=100$ and $f=-0.045$ were selected, and p was confined to $[0, 5.5]$, because larger values cause the hairline to be unstable, and recede past the forehead which authors both believe and hope is not possible. The progress of Androgenetic alopecia is shown in Figure 3

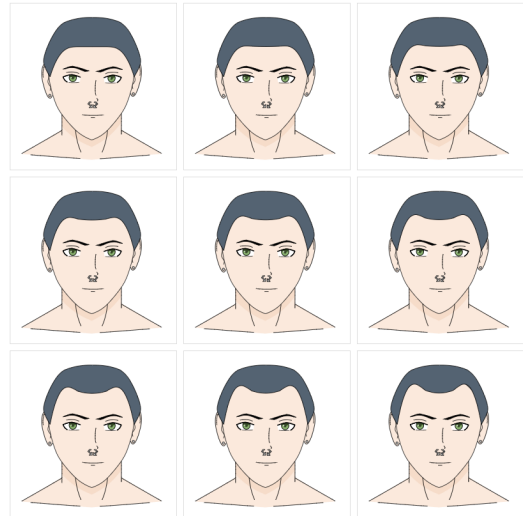


FIG. 3: Receding hairline for p between 0 and 5.5

Furthermore to present easier meltable elements as more bashful, a combination of methods have been used, general blush is adjusted using transparency of paths, and the blush lines start appearing from sides gradually to the center with lower melting points according to a ReLU function.

Our database was based on the first .csv table we found on Google [5], along with data from the website periodictable.com [6].

III. RESULTS

Zooming and looking at faces in detail is advised by authors!

TABLE I: Overview of parameters of the faces, and values which they represent

Face representation	Used functionality	Represented parameter
Smile/Frown	Transforming (scaling)	Electronegativity
Halo	Toggleing	Radioactiveness
Glasses	Toggleing	Syntheticness
Brow density	Coloring	Density
Brow skewness	Transforming (rotation)	Ionization energy
Unibrowness	Transforming (scaling)	Resistivity
Hair color	Coloring	Magnetic Type
Eye color	Coloring	Valence
Receding hairline prominence	Path modification	Young modulus
Ear protrudness	Transforming (scaling)	Poisson ratio
Blush prominence	Coloring	Melting point
Jawline	Toggleing	State of mater in natural conditions
Piercings	Toggleing	Behavior (metal, nonmetal, metalloid)
Symbol	Toggleing	Crystal structure

IV. CASE STUDY

A. Oxygen

We begin our analysis of individual elements with the obvious uke¹: oxygen (Fig. 4). One look at this sad fellow and we immediately know that his electronegativity must be indecently high. The pronounced blush betrays his relatively low melting temperature, while the smooth facial features immediately tell us that he is a gas. The absence of receding hairlines (a rather unpopular trait among young Poles) reveals his tiny Young’s modulus—not to mention the Poisson ratio, which is, in any case, clearly visible from the ears. Not only is he sad, he also seems somewhat lost. The question mark next to his head reveals that this boy has a complicated crystal structure—much like his personality.

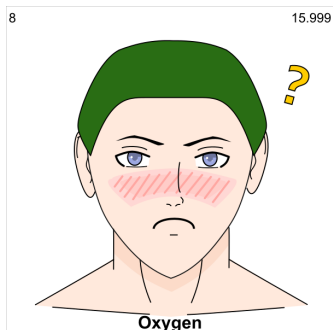


FIG. 4: Chernoff face for oxygen.

B. Iron

Now, for a change, let us talk about a real seme². Take a look at the face of iron (Fig. 5). Do you see that confidence reflected in the earrings? Yes, indeed — it is a metal <3 . The two little stars indicate that we are dealing with a well-organized gentleman: body-centered cubic. Nice. The receding hairline reveals that tensile and compressive tests might have a hard time with him. His over 1500 Celsius degrees melting point steals the most of the blush from his cheeks. What a cold exterior indeed. Iron also stands out for something else — an unusual hair color. Yes, that’s right, he has that special something: magnetic domains characteristic of a ferromagnet. Quite impressive.

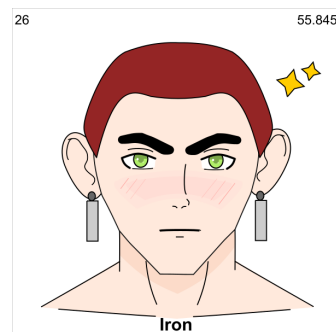


FIG. 5: Chernoff face for iron.

C. Mendelevium

Let us now talk about this clever fellow, Mendelevium (Fig. 6). One of the newer elements (who knows — perhaps the next one will be Chernoffium). The glasses tell us that such brilliant boys can currently be created only in laboratories. At first glance, we see many unknowns: hair in the color “your grandma” reveals that we do not yet know its magnetic nature; the “?” symbol is both a challenge and an opportunity for all those who study the true structure of matter; and the butt-shaped beard hints that even the state of matter remains uncertain — many, many unknowns. But surely he is an angel! You can see the halo! Oh no, my dear — beneath that halo hide horns. This element is radioactive... Yet let us look to the future. Will it not be a beautiful day in the history of humankind when Mendelevium-kun finally receives some hair color? Or when receding hairlines catch up with him? Is that not just as beautiful as a baby’s first teeth coming in??? I love science ;v;

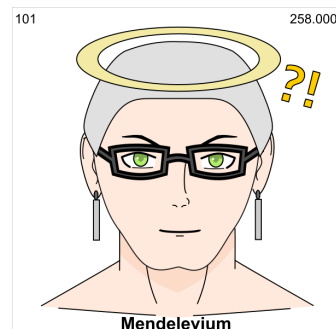


FIG. 6: Chernoff face for mendelevium.

V. BEAUTY CONTEST

To evaluate visual attractiveness of our proposition, a beauty contest was held using Google Forms platform. As beauty is a subjective measure, we used a formula adapted from [8], to accommodate for different tastes,

¹ bottom [7]

² top [7]

or lack of taste. The final score was calculated based on six parameters:

$$\begin{aligned} result = & 0.3 \times \text{handsomness} + 0.3 \times \text{cuteness} \\ & + 0.15 \times \text{speed} + 0.15 \times \text{potential} \\ & + 0.15 \times \text{power} + 0.15 \times \text{durability} \end{aligned}$$

There were ten contestants, five chosen by each of the authors, namely: Cobalt, Wolfram, Chrome, Rubidium, Mercury, Technetium, Thallium, Oxygen, Polonium, Zirconium. Their final placement is as it is shown in the Figure 7.

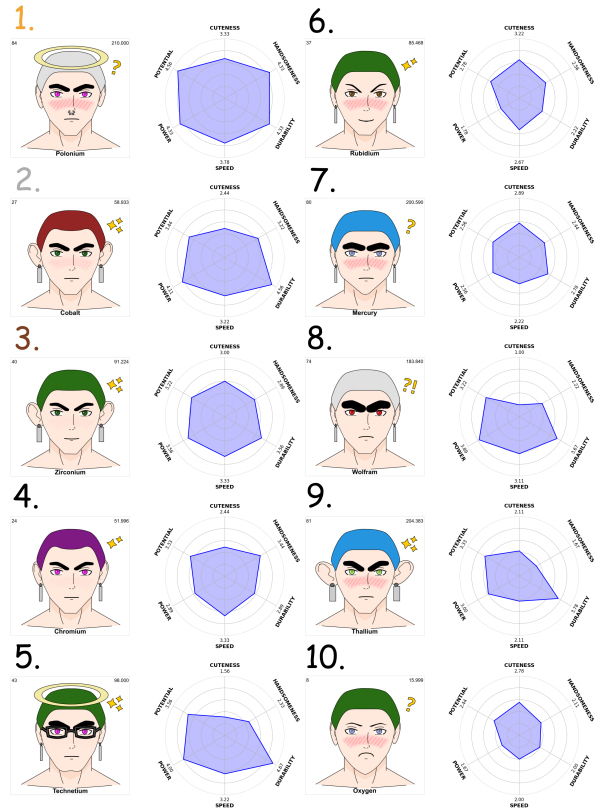


FIG. 7: Statistics acquired in the contest

Polonium claims the first spot. The septum-adorned discovery of Marie Skłodowska-Curie, emanates not only alpha particles, but also excellence, with top-notch results in most of the statistics.

The only quality that was not claimed by Polonium is the durability, where second runner, the ferromagnetic Cobalt, reigns supreme. He might not been deemed cutie by the public, but certainly he is considered quite the powerful handsome fellow.

Interestingly the third place was scored by the Zirconium, who has achieved medium results in all categories.

Just your next-door nice element.

One of the authors was disgruntled learning about poor results for Oxygen and Rubidium, stating that “they could fix them”.

VI. DISCUSSION

Our observations indicate that pattern recognition performance improves when elemental properties are mapped onto facial geometry rather than conventional grid coordinates. Test subjects ($N =$ “a few colleagues we bribed with a Marlboro Gold hyper exclusive cigarette”) were consistently faster in identifying outliers, trends, and approximate value ranges when interacting with representations of the anime-face compared to the classical periodic table.

For instance, Tin-san’s visibly “embarrassed” blush encoding melting point was recalled with higher fidelity than its numerical value alone. While this may alarm purists of inorganic chemistry, it strongly suggests that aesthetic engagement can reinforce cognitive retention without sacrificing informational density. This supports long-standing evidence that the human brain is exceptionally tuned to detect subtle variations in faces.

In conclusion, the fusion of multivariate statistical visualization with stylized anthropomorphic encoding offers a promising direction for high-dimensional scientific communication. Whether adopted in classrooms, laboratories, or as full-wall office décor, the Ultimate Representation of the Periodic Table demonstrates that rigorous science and unapologetic visual flair need not be mutually exclusive.

VII. FUTURE WORK

As we have shown—quite literally—in this article, Chernoff faces constitute a brilliant method for presenting multidimensional data. But are they the ultimate one? One of the authors recently had a dream of an alternative approach: more precise, more complete, perhaps even more perfect. Not Chernoff faces, but Chernoff heads.

When Herman Chernoff originally devised his method of data representation, access to conventional ink printers was far from universal. Times, however, have changed. Today not only does a conventional printer stand in every other household, but in many workplaces one can also find a 3D printer. This technological landscape opens entirely new doors—indeed, a small

paradise of possibilities.

Let us imagine the following situation. A meeting at a financial consulting firm. A long conference table, a presentation glowing on the wall, remarkably handsome gentlemen in sharp suits wearing expressions of grave concentration. One of them presents data on companies competing with their client, but there are many of them—too many. How can attention be maintained? How can one avoid drowning in a flood of numbers and slides?

And then—suddenly!—a cart rolls into the room carrying Chernoff heads.

Each head is parameterized by data: one head, one company. All of them carefully printed on a 3D printer the day before the meeting. Not only do they engage the audience visually, but they also stimulate the sense of touch. Does company XYZ have a massive debt in foreign currency? Instantly you feel the back of the Chernoff head stretching out like the elongated skull of a xenomorph from the Alien films. Missing a patent for its key product? Your fingers detect a conspicuous hole in the temple. Poor employee morale? Impossible to ignore when those colossal bags under the eyes sag dramatically. Perhaps a competitor employs a workforce composed largely of aging specialists? The luxuriant, dignified moustache speaks for itself.

The authors recognize that the present work on the periodic table is already a tremendous contribution to

humanity. For this reason, we do not wish to claim every possible honor for ourselves. Instead, we release the above idea into the intellectual ether. The first brave individual who chooses to pursue it will surely become a hero.

CONFLICT OF INTEREST

The authors declare the following potential conflicts of interest. The following individuals were involved in circumstances that could be perceived as influencing the preparation or interpretation of this work: K. Ł. Detonatorewicz, P. C. Czechrepublikiewicz, J. H. Anusewicz, D. A. Weed, D. Tusk, W. O. Juliwny.

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The authors further note that approval from the relevant bioethics committee was not obtained for the research described in this work.

All the computing was done on my computer.

At the conclusion of the experiment, eleven cigarettes remained in the aforementioned pack of Marlboro Gold.

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